Checkpoint C: Database Implementation

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Abstract

The purpose of this paper is to outline the construction and possible uses of a knowledge base concerning public companies in the healthcare sector. This outline is provided in three main sections. These are a literature review, a section entitled methods, and a results section. In the literature review the researchers detail related published works. The methods section presents the project approach for the database in addition to the eight steps outlined by Chakrabarti et al. (1999). Finally, the results section details outcomes for some of the work completed thus far.

Introduction

Traditional methods of evaluating the future performance potential of stocks were mainly computational. Over time various other methods have been proposed including analyzing sentiment of publicly available information. One of the earliest and most influential papers challenging the status quo was that by Cootner (1962). He challenged the prevailing theory of the trajectory of stock prices being purely a random walk forecastable by precise calculations. Cootner (1962), instead proposed that this random walk wavered between less precise price barriers. This wavering was driven by professional traders not utilizing the same methodology, and non-professional traders assessing value with less precise measures (Cootner, 1962).

The work by Cootner (1962) laid the foundation that has since been expanded on. This began as early as the 1970's with work by Zweig (1973) to analyze closed-end funds as an indicator of investor sentiment. In this work, the researchers present a database and plan to expand upon this area with the intent of analyzing sentiment related to specific stocks present in various focused news articles. This will provide the ability to forecast whether a particular stock would make a good investment. This begins with a literature review of some of the more recent efforts of other researchers in this area followed by a presentation of the planned methods.

Literature Review

Various other researchers have investigated the merits of utilizing sentiment found in online text documents for forecasting stock performance. Some of these studies such as that by Feuerriegel and Gordon (2018) have utilized company and regulatory reports. Additionally, the works presented by Schumaker et al. (2012), Wüthrich et al. (1998), and Wu et al. (2021) have utilized news articles. Finally, some such studies have also utilized social media and comment boards. Examples of this final type are works by Deng, Sinha, and Zhao (2017), Li and Li (2013), and Bollen, Mao, and Zeng (2011). Following is a review of one study from each of these three approaches.

The research by Feuerriegel and Gordon (2018) analyzed sentiment in the text of 75,927 ad hoc regulatory announcements in both English and German. Researchers in this study utilized several different machine learning techniques to analyze this text data including "lasso, ridge regression, elastic net, gradient boosting, principal component regression, and random forest" (Feuerriegel and Gordon, 2018). They then used the results of these analyses to predict the performance of the German stock indexes DAX, CDAX, and STOXX Europe 600. The results of utilizing sentiment analysis were compared to results from a linear autoregressive model utilizing historic time series price data (Feuerriegel and Gordon, 2018). Feuerriegel and Gordon

(2018) found their methods utilizing sentiment analysis were able to outperform established linear models when forecasting future stock prices.

Schumaker et al. (2012) chose to analyze sentiment in news articles for their research. The researchers in this article utilized the existing stock prediction tool AZFinText and augmented the results from that system utilizing the OpinionFinder tool. The combined results were then used as instructions for a trading engine to make a trade 20 minutes after each prediction was made. The researchers indicated that their approach netted returns from those trades between 2.4 and 3.3 percent (Schumaker et al., 2012).

In their research Bollen, Mao, and Zeng (2011), utilized the tools OpinionFinder and Google-Profile of Mood States to measure public reactions around the 2008 election. The researchers then compared the results of this analysis to changes in the value of the Dow Jones Industrial Average (DJIA). The researchers found that simply comparing the sentiment results to trends in the DJIA did not seem to reveal much correlation. However, they then trained a Fuzzy Neural Network with their results organized in a time series along with historical DJIA values. They claimed the results of the neural network showed a higher degree of accuracy compared to their initial attempts (Bollen, Mao, and Zeng 2011).

Methods

The healthcare sector is incredibly dynamic and is responsible for human progress through the development of cutting-edge technologies. This sector is also distinctive in that it is less sensitive to broader economic fluctuations compared to most industries (Correa 2024). Given these characteristics, the healthcare sector presents a unique opportunity for focused analysis of its leading companies in the market.

The intent of the researcher's database is to provide the information necessary for an analysis of a stock's future performance. So far, data from Wikipedia and news articles regarding the sector and its top 10 organizations by market cap have been included. The intended primary vehicle of analysis is sentiment analysis. This will be layered on top of the database either as an analysis of its contents or of the gathered articles and the results added to the structure built thus far.

The intended user of this tool is likely a financial professional such as a fund manager or financial analyst. However, it is not limited to only professionals, but anyone looking to invest, such as retail investors. Successful implementation of the knowledge base relied on prerequisite steps that included the identification of appropriate web sources, the development of multiple website specific web crawlers using the Python programming language to perform web scraping with the scrapy package, and the development of a database schema where the information will be housed and built with the graph-relational database Gel.

The knowledge domain needed to encapsulate the healthcare sector for effective sentiment analysis. The healthcare sector is vast, not only in the range of company sizes, but also in the diversity of its subsectors. For this reason, the top ten healthcare companies by market capitalization are the focus of the crawlers, as they are the primary drivers of the market (Kavout 2025). Web sources chosen for the analysis are centered around these ten companies.

As outlined by Chakrabarti et al. (1999), the researchers for this project researched possible sources whose websites can be crawled for relevant information. This included both manual inspection of sample web pages and web crawler testing. The goal being to verify data availability, assess page structure, and identify potential challenges such as dynamic content or potential pay wall issues.

The team's original intention was to incorporate text data from social media postings relating specifically to the 10 target corporations. In performing further research as to the crawlability of various social media and news websites for potential text data, multiple websites appeared to be actively restricting the activities of website crawlers. While some news websites appeared to restrict crawler activity, the social media websites were even more stringent. We therefore refocused our efforts toward news articles. Those efforts resulted in three successful web crawlers and one API thus far. The first of these is for Wikipedia. This first source provides background information on the ten companies such as name, ticker symbol, market capitalization, location and description. Two web crawlers have also been successfully created for news articles from CNBC and Investopedia. The CNBC web crawler obtains text data from a list of news articles relating to the healthcare sector. The Investopedia web crawler works in two parts. First, it gathers links to news article search results for each of the 10 selected companies. The webcrawler then proceeds to pull text data from all of the identified news articles.

Chakrabarti et al. (1999) discusses the importance of incorporating a distiller and classifier into a full-blown focused crawler. For the crawlers in this project, the researchers are utilizing the news site's native search function as the means of distillation. Three approaches have been utilized for this. In the first method, links from the search results are gathered manually and then incorporated as an iterative list in the web crawler instructions. The second approach automates the process to gather links for news articles. In this approach the researchers first identify the html class tag relating to links for news articles in the search results. The crawler is then programed to identify this class in the target page's html and gather the associated links which it then uses an iterative list for data extraction. The third method utilizes an API to fetch data from top new sources that match defined tags such as "healthcare".

The process of defining a schema for the database is necessary for laying a foundation for data storage once text data is obtained through the crawler activities. The type of database used for this study is a graph-relational database built using Gel. This schema combines relational tables (objects) with graph-like relationships (links), making it ideal for modeling our healthcare sector effectively.

Once the data is properly parsed and cleaned it was fed into the database. Nodes include company, sentiment source, and financial metrics. Properties stored within the company nodes include name, ticker, market cap, and description. This serves as the central node, connecting to sentiment sources, financial metrics, and sectors for sentiment and financial analysis. Sentiment source nodes will have properties such as source type, content, publish date, sentiment score, and url. These nodes will connect companies to capture sentiment data, assisting with the analysis of public perceptions. The financial metric nodes have the properties metric type, metric value and date. These nodes will link to companies to track financial trends over time.

Edges link the relationships between nodes that will resemble: Company -to- Sentiment Source (Multi-Link), Sentiment Source -to- Company (Multi-Link, Inverse), Company -to- Financial Metrics (Multi-Link), Financial Metrics -to- Company (Single Link, Inverse).

With this initial structure in place, the final application will serve to build up a knowledge base to expose users to the sentiment within the healthcare sector. The intended purpose of this application is for information retrieval within the knowledge base. Additionally, this market sentiment application aids in the decision-making process for trading stocks by offering the user recommendations of buy or sell points in the market. The completion of these methods outlined will ensure a complete foundation for the subsequent analysis, enabling an informed interpretation of the sector's underlying forces and market drivers.

The knowledge graph was constructed using a preprocessing pipeline. The pipeline was set up to first clean and normalize jsonlines files that housed the text data obtained from the web through crawlers and APIs. Next, entities and relationships were extracted from the cleaned text script using the Python library spaCy. SpaCy's NLP capabilities were used to extract structured information from the text data obtained from the web crawlers. Our approach processed the cleaned_text.jsonl file that was cleaned and normalized previously. Using the en_core_web_sm model from spaCy, the pipeline applies named entity recognition (NER) to identify entities labeled as "org" (organizations), "product", "gpe" (geopolitical entities), and person. A rule-based matching approach scanned for known company names in the text, mapping them to ticker symbols for graph linkage. This was added to supplement spaCy's NER to ensure coverage of company mentions and assisting with stock price ingestions. For each sentence, the script parses syntactic dependencies to extract subject-verb-object triples. The relations were kept only if their subjects and objects were found using dependency labels like "subj" and "obj." The verb, which acts as the relation's predicate, is standardized by lemmatizing it from the subject. The data is saved in a json lines file that is then used for the gel mapping script.

The extracted entities, relations, and metadata from news articles are mapped into a format suitable for ingestion into the Gel database. This script processes articles to identify and aggregate unique entities based on their text and label. It attempts to associate each article with a known company ticker by matching company names from a dictionary or by substring search within the article's text. For each relation found in the articles, it records the relation along with metadata such as the associated company ticker, publication date, URL, title, and content. The script outputs two files: one containing all unique entities and another containing all relations with their associated data. The purpose of this mapping is to ensure that remaining ingestion scripts can insert entity and relation data into the Gel database, linking them to relevant companies.

The companies and their associated tickers were loaded into the database using a separate python script. The rationale for this approach stems from the fact that we want our scope to stay within the top ten healthcare companies. Adding them in manually ensures that our focus stays within the predefined project scope. Once this was performed the next step involved ingesting the news article data into the database.

The next process in the pipeline involved the actual ingestion of preprocessed json data into the Gel database to match the predefined schema in appendix 1. The code utilizes coroutines using the asyncio standard library in Python and Gel's async client to run queries. The script handles three types of insertions obtained from former steps in the process, named entities, subject-verb-object relations, and news articles. The script first inserts basic named entities from a json lines file into the “Entity” type. Next, it processes relation triples and associates each relation with a “Company” using its corresponding stock ticker. The last insertion takes the “NewsArticle” entries, and checks for duplicate URLs to maintain uniqueness before adding them into the database. It also links each article to relevant Company nodes through the mentions_companies link, using extracted tickers.

The very last piece of the data ingestion process involved adding in stock pricing data for each company dating back two years. The yfinance Python library was used to fetch the data for each company of interest and for a two year time period. That data was called in a stocks processing script that first extracts the fields, date, close, open, high, low, and volume. It performed a lookup to ensure the associated “Company” exists in the graph, using the stock ticker as the key. Once found, it inserted a “StockPrice” record linked to that company with the associated data. Upon the successful execution of this data insertion process, the system will provide a printout with the total number of records that were successfully added to the database. This output serves as a critical verification point, confirming the integrity and completeness of the data transfer.

Results

Crawlers for Wikipedia, CNBC, Investopedia, and NewsAPI have successfully retrieved healthcare sector data which has been added to the database. Figure 1 illustrates a news article that was scraped from CNBC. The figure shows that the URL, source, title, tags and article content were successfully obtained. While success in scraping was obtained using Wikipedia and CNBC, Yahoo Finance was met with challenges. When the crawler was set up to scrape company news data from Yahoo Finance a 429 restricted error was observed. This error is typical of too many requests to the server, but adjustments made to the request delay did not correct the error. Further review suggested that Yahoo Finance has changed policies with how web crawlers are allowed to interact with the content. This is likely the cause for the difficulties observed with the process.

The NewsAPI approach was successful. With this approach the data obtained is from various reputable news sources writing about the healthcare sector, but the content is limited to 100 words. Given that many articles contain extraneous content not required for sentiment


```
{"url": "https://www.cnbc.com/2025/05/01/
eli-lilly-ceo-david-ricks-trump-pharmaceutical-tariffs.html", "source":
"cnbc", "title": "Eli Lilly CEO says company can help with national
security concerns around pharma", "text": [], "author": null,
"timestamp": null, "tags": ["health", "science", "news", "cnbc"],
"content": "In this article  CEO Dave Ricks on Thursday said the
drugmaker can help \"respond\" to national security concerns around
cheaper essential medicines as   loom.  The Trump administration has
opened a Section 232 investigation into how importing certain drugs into
the U.S. affects national security – a move widely seen as a prelude to
initiating tariffs on pharmaceuticals. It is unclear what those levies
will look like and whether they will target branded or older generic
drugs, the latter of which are largely made overseas in countries like
India and China.  \"Bringing that capacity back, so in case of
emergency, we have the stock, we have the supply – that's a valid thing,
\" Ricks said in an interview with CNBC, referring to those older drugs.
He spoke after Eli Lilly  , which did not include estimated effects of
the potential pharmaceutical tariffs. He said national security concerns
around those medications are \"valid.\" But he added: \"Do I think
tariffs are the answer to that? I'm not so sure personally.\" \"We would
be happy to talk to this administration or national security people
about how we could respond to such a crisis,\" he said. \"We have
capacities to bring to bear there, and we're happy to help the country
if we're in need.\" Older generic drugs account for about  prescribed in
the U.S. Many are critical for hospital care, including antibiotics and
```


Figure 1: Sample news article obtained from CNBC and stored in JSON Lines format.

classification, we are confident that the available material is sufficient to accurately determine the sentiment. Figure 2 provides a sample of some of the news that api is fetching. Additionally, efforts made to obtain news information from Investopedia have been conducted successfully. With this approach we scraped 160 recent news articles to date. This collection includes articles for each of the top 10 companies. A sample of one of the articles obtained with the Investopedia crawler is shown in figure 3.


```
{"source": {"id": null, "name": "Yahoo Entertainment"}, "author": "Michael Erman", "title":
"Bristol Myers posts better-than-expected quarterly revenue on strong cancer drug sales",
"description": "(Reuters) -Bristol Myers Squibb reported better-than-expected first-quarter revenue on
Thursday and raised its full-year forecast due to growth from its...",
"url": "https://ca.finance.yahoo.com/news/bristol-myers-posts-better-expected-110305143.html",
"urlToImage": "https://media.zenfs.com/en/reuters.com/506e70cbb31e951dd227ed295c47ce7e",
"publishedAt": "2025-04-24T11:03:05Z", "content":
"By Michael Erman\r\n(Reuters) -Bristol Myers Squibb reported better-than-expected first-quarter
revenue on Thursday and raised its full-year forecast due to growth from its portfolio of drugs that spur…
[+1980 chars]", "tags": ["healthcare"]}
```


Figure 2: A sample of healthcare news obtained using NewsAPI.org


```
2    {"url": "https://www.investopedia.com/
     what-to-expect-in-the-markets-this-week-april-28-2025-11721321", "title": "What To
     Expect in the Markets This Week", "text": "Key TakeawaysPresident Donald Trump's
     100th day in office comes on Wednesday.Apple, Amazon, Microsoft, Meta Platforms,
     ExxonMobil, Coca-Cola and McDonald\u2019s are among the firms scheduled to release
     quarterly results in a packed week of corporate earnings.The Federal Reserve will
     get the April jobs report and key inflation data this week as Trump has reiterated
     his calls for the central bank to cut interest rates.Investors will also be
     watching out for first-quarter GDP data, the latest consumer confidence report, a
     trade balance update and housing market reports. Major corporate earnings, April
     jobs data and the latest inflation report are on tap for investors this week.
```


Figure 3: A sample section of a news article obtained from Investopedia.

With confirmation of viable data availability for the database, the focus shifted to defining the schema for the new database. Appendix A illustrates the schema definition language (SDL) code for Gel. This demonstrates the approach used to build the database. Wikipedia sources filled in the company background data such as location, founding year, products produced, and acquisitions etc. CNBC, NewsAPI and Investopedia sources provide current happenings in the market and will be the bulk of text used for sentiment analysis. This data will be housed in the SentimentSource type outlined in the schema.

With the entities and relations from the crawl and API work loaded into the schema for the knowledgebase, this information can now be queried to answer questions related to the healthcare sector. Covered here are a few examples. First, a business user may want to see recent news headlines related to a specific stock. This information could be pulled using the following query:

Sample Query A:

“select NewsArticle{title} filter .mentions_companies.stock_ticker = 'LLY' order by .publication_date desc limit 10”

Second, a trader may need to see recent stock quotes which can be performed with the query:

Sample Query B:

“select StockPrice {date, close} filter .company.stock_ticker ='LLY' order by .date desc limit 5”

Third, a researcher may wish to see how often certain organizations are appearing in the news. This could be accomplished with the following query:

Sample Query C:

“select Company {name, mention_count := count(.<mentions_companies[is NewsArticle])} order by .mention_count desc limit 5”

Appendix B contains the results printed in the terminal that are obtained from each sample query outlined above.

Conclusions

The knowledge base constructed lays the groundwork for analysis on the future prospects of a given stock. Thus far, general information about the stock, stock quotes, product information, and news articles have been incorporated. This general information allows an investor to learn general information about an organization, check recent events concerning the organization, and recent stock price trends.

The schema has also been constructed to accommodate the inclusion of sentiment scores. The researchers are working toward including a sentiment analysis score for each article. Current efforts focus on utilizing BERT for sentiment analysis. The plan is to utilize a pre-trained model for BERT and layer in additional model training using a subset of the articles gathered. This subset has been hand labeled with sentiment labels according to whether the article is overall optimistic or pessimistic on the market outlook. Sentiment analysis with NLTK or SPACY was also considered but the researchers decided it may be more accurate given the subject matter to utilize a tool that allowed for training. Depending on the results with BERT, these tools may be revisited. The combination of general stock information and performance combined with sentiment from related articles would allow an investor to make informed decisions.

Appendix A: Schema Definition Language Outlined for the Proposed Gel Database

The following Schema Definition Language (SDL) code defines the structure of the Gel database designed for a knowledge base focused on the healthcare sector.

```
module default {

    type Company {
  required name: str;
  required stock_ticker: str { constraint exclusive; }
  headquarters: str;
  founding_year: int32;
  revenue: float64;
  employees: int32;
  market_cap: float64;
  industry: str;
    }

    type StockPrice {
  required date: cal::local_date;
  required close: float64;
  open: float64;
  high: float64;
  low: float64;
  volume: int64;
  required link company: Company;
  constraint exclusive on ((.date, .company));
    }

    type Product {
  required name: str;
  type: str;
  approval_date: cal::local_date;
  revenue: float64;
  link manufacturer: Company;
    }

    type NewsArticle {
  required url: str { constraint exclusive; }
  required title: str;
  publication_date: datetime; # Changed to datetime for precision
```

```
    source: str;
    author: str;
    summary: str;
    content: str;
    tags: array<str>;
    category: str;
    metadata: json;
    sentiment_score: float32; # For sentiment analysis
    multi link mentions_companies: Company;
    multi link mentions_products: Product;
    }

}
```

Appendix B: Sample query results

Sample Query A:

{"title": "eli lilly lly option traders confident after earnings"}
{"title": "eli lilly stock drops as lowered profit outlook outweighs solid q results"}
{"title": "top stock movers now ab inbev molson coors eli lilly and more"}
{"title": "eli lilly lly option traders prepped for earnings beat"}
{"title": "top stock movers now unitedhealth nvidia eli lilly and more"}
{"title": "novo nordisk stock drops on eli lilly oral weightloss drug success downgrade"}
{"title": "eli lilly stock soars on oral weightloss drug trial results"}
{"title": "top stock movers now delta apple eli lilly and more"}
{"title": "hims hers stock surges as telehealth platform adds eli lilly weightloss drugs"}
{"title": "cvs to boost access to novo nordisks wegovy for patients on its drug plans"}

Sample Query B:

{"date": "2025-05-23", "close": 713.7100219726562}
{"date": "2025-05-22", "close": 715.2000122070312}
{"date": "2025-05-21", "close": 724.9500122070312}
{"date": "2025-05-20", "close": 747.010009765625}
{"date": "2025-05-19", "close": 755.1099853515625}

Sample Query C:

{"name": "Eli Lilly", "mention_count": 20}
{"name": "Merck & Co", "mention_count": 10}
{"name": "Abbott Labs", "mention_count": 9}
{"name": "Amgen", "mention_count": 8}
{"name": "Thermo Fisher Sci", "mention_count": 7}

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